

Opinion Pooling III: Desirable Features of Pooling Strategies

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Formal Pooling Axioms

Last time we discussed:

- Group rationality and justified group belief.
- Inflationary vs. deflationary accounts
- Group responsibility and its connection to pooling.

Today: What formal properties should pooling functions satisfy? What are the trade-offs?

What makes a pooling strategy good. Elkin and Pettigrew use what they call an **axiomatic approach**.

- Axioms are desirable formal properties we want pooling functions to satisfy.
- We've seen this before with Arrow; axioms are things like No Dictators, or Pareto-optimality.
- Different axioms capture different intuitions about what pooling should do.
- Will there be some impossibility results incoming?
- Yes, of course there will, there always are around here.

A pooling function takes individual credences and outputs a group credence.

A credence function measures how confident each person is in each proposition.

The values are between 0 and 1, with higher numbers representing greater confidence.

There are lots of arguments that credence functions should be **probability functions**. These arguments are not entirely uncontroversial, but they are very widely accepted, especially outside philosophy, and we will accept them here.

A function Pr is a probability function (as we're understanding them - this isn't exactly how mathematicians would understand them), satisfying these constraints.

1. For all A , $Pr(A) \in [0, 1]$
2. If $A \vdash B$, then $Pr(A) \geq Pr(B)$.
3. If $\vdash \neg(A \wedge B)$ then $Pr(A \vee B) = Pr(A) + Pr(B)$.

We're not covering this, but Elkin and Pettigrew talk about it a lot, so I wanted to flag it here.

Some people, **including many people in UM philosophy** think it's better to model people with sets of probability functions than with individual probability functions.

The person thinks A is more likely than B just in case $Pr(A) > Pr(B)$ for every member of the set.

With some extra bells and whistles, this is called the imprecise credence approach.

The bits of Elkin and Pettigrew we won't cover, starting with section 4.4, are largely about applying this approach to pooling.

I could talk about this at length. I literally did a PhD thesis on it back when you had to write them on scrolls.

But I think it's a distraction here; the precise case, where individual credences are individual probability functions is (a) much more widespread across different fields, and (b) hard enough.

We're looking for a pooling function F from individual credences to group credences.

- **Inputs:** Each person i has a credence function Pr_i over some propositions.
- **Output:** The pooling function F produces a group credence function $Pr_G = F(Pr_1, Pr_2, \dots, Pr_n)$.
- **Goal:** F should satisfy various desirable properties.

The question: Which properties are most important for which contexts?

Basic Desiderata

Unanimity: If all individuals assign the same credence to a proposition, the group should too.

Formally: If $Pr_i(A) = x$ for all individuals i , then $Pr_G(A) = x$.

- This seems really—if everyone agrees, that should be the group view.
- Violation would mean the group disagrees with all its members.
- Sometimes called **Pareto principle** in social choice theory.

Consider a medical committee where all experts agree a treatment is 90% effective.

- If the pooling function gives the group credence of 70%, something has gone wrong.
- The group belief no longer represents the consensus that actually exists.
- This could lead to poor decisions based on views no one actually holds.

Good news: Many standard pooling methods satisfy unanimity.

Consider a case where two detectives are investigating a murder.

- One interviews all the suspects and witnesses, and ends up 90% sure that the butler did it.
- The other looks carefully at the physical evidence, and ends up 90% sure that the butler did it.
- Let's say we know what each detective thinks, and nothing else; and the detectives don't know each other.
- How should we pool their credences?

Maybe more than 90%? After all, both the witnesses and the physical evidence point strongly in the same direction.

Eventwise Independence: The group's credence in A should depend only on individual credences in A , not on credences in other propositions.

Formally: If $Pr_i(A) = Pr'_i(A)$ for all i , then
$$F(Pr_1, \dots, Pr_n)(A) = F(Pr'_1, \dots, Pr'_n)(A).$$

This seems reasonable: why should our credences about *the weather tomorrow* affect how we pool opinions about *election outcomes*?

The Intuition Behind Eventwise Independence

Imagine pooling credences about whether it will rain tomorrow.

- Alice: 70% chance of rain
- Bob: 40% chance of rain
- The pooled credence should depend only on these numbers.
- It shouldn't matter what Alice and Bob think about the stock market, or anything else.

This is a **separability** condition—different propositions can be pooled independently. It's like Arrow's **Independence of Irrelevant Alternatives** condition.

The Problem with Eventwise Independence

But this axiom, like Arrow's, turns out to be very restrictive.

- It's incompatible with some intuitive pooling methods.
- It can prevent the pooling function from respecting probabilistic structure.
- We'll see this creates tensions with other desirable properties.

The big thing is that propositions aren't always that separate. Should our individual opinions about $A \wedge B$ be relevant to the pooled opinion on A ? Maybe!

Avoiding Extremes

Non-Dictatorship: No single individual should always determine the group credence regardless of others' opinions.

Formally: There should be no individual i such that $Pr_G = Pr_i$ for all profiles.

Why this matters:

- Groups exist to aggregate multiple perspectives.
- A dictator rule makes the rest of the group irrelevant.
- It isn't really a **group** opinion if that's how things go.

When Might Dictatorship Be Tempting?

Suppose we have a group with one genuine expert and several novices.

- Shouldn't we just use the expert's credence?
- Maybe for *that particular question*, but not as a general rule.
- And again, things like conjunctions make life really complicated.

Figuring out the group credence in $A \wedge B$, when one person is an expert on A alone, turns out to be *really hard*.

Non-dictatorship is consistent with **differential weighting**.

Bounded Pooling: The group credence should lie between the minimum and maximum individual credences.

Formally: $\min_i P_i(A) \leq P_G(A) \leq \max_i P_i(A)$.

This is sometimes called being in the **convex hull** of individual opinions.

If Alice thinks rain is 30% likely and Bob thinks it's 80% likely:

- The group shouldn't say 90% (more extreme than anyone).
- The group shouldn't say 20% (more extreme than anyone).
- The pooled view should be a compromise, not an extrapolation.

Most pooling functions satisfy this, but it's worth stating explicitly. And again, the detective case from earlier complicates things.

Pooling and Learning

Imagine the following situation. We're a group of scientists, and we all have opinions on some hypothesis H . We're in the middle of running some experiment E , and while E won't settle H , the results will be relevant.

Here's a puzzle: When should updating happen relative to pooling?

Two scenarios:

1. **Pool then update:** Combine credences, then update on experiment E .
2. **Update then pool:** Each person updates on E , then pool the updated credences.

Should these give the same result?

Update Tolerance: The order of pooling and updating shouldn't matter.

Formally: $F(Pr_1(\cdot|E), \dots, Pr_n(\cdot|E)) = F(Pr_1, \dots, Pr_n)(\cdot|E)$.

- This says: “pool-then-update” = “update-then-pool”
- It ensures learning and pooling don't interfere with each other.
- Violations mean the group's response to evidence depends on procedural choices.

Imagine a research team receives new data.

- If they pool first, then update on the data as a group, they get one answer.
- If each person updates individually, then they pool, they might get a different answer.
- Without commutativity, the group's belief depends on meeting logistics, which seems bad.

Unfortunately, commutativity is very demanding.

- It rules out many otherwise attractive pooling methods.
- Combined with other axioms, it leads to impossibility results.

We'll see which pooling methods preserve this property.

This principle goes by any number of different names:

- Commutativity
- Update Tolerance
- External Bayesianity

I don't even know how to pronounce that last one, but it's the one the book uses.

Independence Preservation

Probabilistic Independence: Propositions A and B are independent if $Pr(A \wedge B) = Pr(A) \cdot Pr(B)$.

Question: If individuals all judge A and B independent, should the group?

Independence Preservation: If $Pr_i(A \wedge B) = Pr_i(A) \cdot Pr_i(B)$ for all i , then $Pr_G(A \wedge B) = Pr_G(A) \cdot Pr_G(B)$.

Independence judgments reflect important structural beliefs.

- “Coin flip outcomes are independent” is a structural claim.
- If everyone agrees on this structure, pooling should preserve it.
- Otherwise, we introduce dependencies that no one believes exist.

This connects to justified group belief: the group should respect shared evidential structure.

Independence preservation is **very hard to achieve**.

- Most pooling methods violate it.
- It conflicts with other desirable axioms.
- This is one source of impossibility results in pooling theory.

We'll see that different pooling strategies make different trade-offs here.

Practical Rationality Constraints

Here's a pragmatic worry: Could a pooling method lead to guaranteed losses?

No Sure Loss: The group shouldn't accept bets that guarantee losses no matter what happens.

- If the group is incoherent (violates probability axioms), it can be Dutch-Booked.
- But even coherent pooling can create problems with sequential decisions.
- The group might regret its earlier choices.

Suppose the group makes a decision based on pooled credences at time 1.

Later, at time 2, the group pools again (maybe with new evidence).

No Regret: The group shouldn't systematically wish it had acted differently at time 1.

- This is about diachronic consistency.
- The group's future self shouldn't predictably undermine its past decisions.
- Violations suggest the pooling method is unstable over time.

This connects to our earlier discussion of group responsibility.

- If a pooling method guarantees regret, who's responsible for bad outcomes?
- Is it fair to hold the group accountable for predictably bad decisions?
- The choice of pooling method carries moral weight.

This motivates careful attention to which axioms promote practical rationality.

The Impossibility Landscape

Here's the bad news: No pooling function satisfies all desirable axioms.

Classical impossibility results:

- No linear rule satisfies External Bayesianity plus Non-Dictatorship.
- Only linear rules satisfy Eventwise Independence and Local Unanimity Preservation.
- Putting these together: nothing satisfies External Bayesianity, Non-Dictatorship, Eventwise Independence, and Local Unanimity Preservation.
- There are *many* results like this.

We face genuine trade-offs:

- Satisfying some axioms requires violating others.
- Different contexts may prioritize different axioms.
- There's no one-size-fits-all pooling method.

The rest of the book explores which trade-offs make sense for which purposes.

Despite impossibilities, the axiomatic approach is valuable:

- It clarifies exactly what we're giving up with each choice.
- It reveals hidden connections between properties.
- It helps us choose pooling methods for specific contexts.
- Medical decisions might prioritize different axioms than democratic votes.

Looking Ahead

Having surveyed the axioms, we can now:

- Examine specific pooling functions (linear, geometric, multiplicative).
- See which axioms each function satisfies or violates.
- Connect the formal results to our earlier philosophical discussions.
- Ask: Which trade-offs are justified by epistemic or practical goals?

Next lecture: We'll examine specific pooling strategies:

- **Linear pooling:** Taking weighted averages.
- **Geometric pooling:** Multiplicative aggregation.
- **Imprecise pooling:** Representing group opinion as a set of credences.

We'll see how each performs on the axioms we've discussed today.